# Index construction and MapReduce

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(partially based on the slides from the Stanford course on IR)

March 23, 2026

# MapReduce: an example of NoSQL data management

- Data model: key-value pairs
- Originated at Google
- Need for large scale parallellism to create an index for the WorldWideWeb
- MapReduce facilitates massive parallellism on large amounts of commodity hardware
- It provides the programmer with a generic interface for parallel programming ...
- ... and supports transparency with respect to low level implementation issues, especially robustness

# Example: an index for text collections

In general for document collections:

*INPUT: a term*
*OUTPUT: the corresponding postings list, i.e. all documents in the text collection containing this term*

Specifically for the Web:

*INPUT: a term*
*OUTPUT: the corresponding postings list, i.e. all URLs of web pages containing this term*

# Creating postings lists

*Input :* document collection <docid, text>

< 2013, "de dag die je wist dat zou komen is eindelijk hier" >
< 1980, "de do do do, de da da da" >
< 1971, "jaren komen en jaren gaan" >
< 1994, "we komen en we gaan" >

*Output :* a set of *postings lists* for this collection of documents

...
<"dag", [2013] >
<"de", [1980, 2013] >
<"die", [2013] >
<"do", [1980] >
<"en", [1971, 1994] >
<"gaan", [1971, 1994] >
...
<"komen", [1971, 1994, 2013] >
...

# Index for text collections

INPUT: a term
OUTPUT: the corresponding postings list, i.e. all occurences of this term in the text collection containing this term.

Two steps:

- Use a tree structure (B-tree, suffix tree) that connects a term to the corresponding postings list
- Return the postings list

# Postings list sorted

Query = $term_1$ AND $term_2$

1. locate postings list $p_1$ for $term_1$
2. locate postings list $p_2$ for $term_2$
3. calculate the intersection of $p_1$ and $p_2$ by list merging

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $term_1 \Longrightarrow$ | 1 | 3 | 7 | **11** | 37 | **44** | **58** | 112 | ... |
| $term_2 \Longrightarrow$ | 2 | 4 | **11** | 25 | **44** | 54 | 55 | **58** | ... |

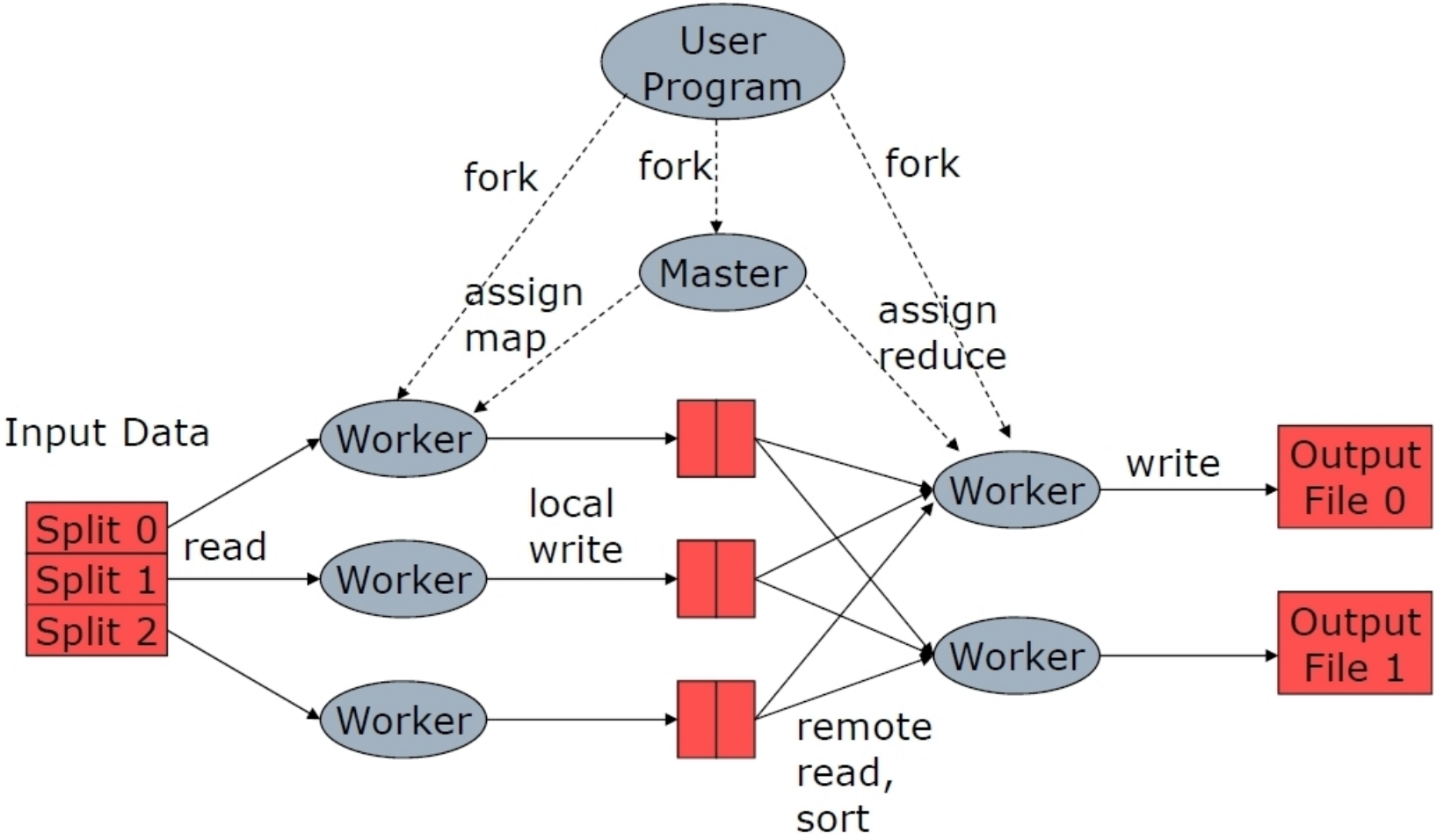
# MapReduce

- Two sets of machines involved in parallel processing: *Map* workers and *Reduce* workers
- Robustness by replication of data in file system
- Generic, based on *Map* and *Reduce (Fold)* from functional programming
- Several implementations, Hadoop is the most well known

# MapReduce: the Map

- Basic data structure is key-value pair $< k, v >$
- Input is split into disjoint chunks, containing collections of key value pairs
- Each Map worker works autonomous from other map workers ("shared nothing")
- Each Map worker scans it's own input chunk once
- Each Map worker does one uniform calculation on each key-value pair
- The output of each Map worker is a set of key-value pairs: zero, one or more
- The structure of the resulting key-value pairs is generally different from the input pairs
- The output results of all Map workers are collected for further processing in the Reduce phase

# MapReduce computing



# MapReduce: the Reduce

- The output results of all Map workers are grouped on the key values and assigned to the reduce workers
- All related key value pairs will be processed by one Reduce worker
- Each Reduce worker works autonomous from other Reduce workers (shared nothing)
- The output results of all Reduce workers together are the result of the calculation

# MapReduce example

Example: *word count*

Input: a collection of documents
Output: the words in the documents with their frequency

- Map $< docid, text >$:
  for each word $w$ in $text$
  $emit(< w, 1 >)$;
- Reduce $< w, vlist >$:
  ...

# MapReduce example

*Input to Map-workers:*

$<$ 2013, "de dag die je wist dat zou komen is eindelijk hier" $>$
$<$ 1980, "de do do do, de da da da" $>$
$<$ 1971, "jaren komen en jaren gaan" $>$
$<$ 1994, "we komen en we gaan" $>$

*Output from Map workers:*

$<$"de", 1 $>$
$<$"dag", 1 $>$
$<$"die", 1 $>$
...
$<$"gaan", 1 $>$

# MapReduce example

... then comes the invisible step ...

... which could be characterized as a "GROUP BY key" ...

# MapReduce example

*Input to Reduce-workers:*

$<$"de", [1] $>$

...

$<$"komen", [1, 1, 1] $>$

...

$<$"gaan", [1, 1] $>$

...

---

*Output:*

$<$"de", 1 $>$

...

$<$"komen", 3 $>$

...

$<$"gaan", 2 $>$

...

# MapReduce example

Example: *word count*

Input: a collection of documents
Output: the words in the documents with their frequency

- Map $< docid, text >$:
  for each word $w$ in $text$
  $emit(< w, 1 >)$;
- Reduce $< w, vlist >$:
  int $sum = 0$;
  for each $v$ in $vlist$
  $sum++$;
  $emit(< w, sum >)$;

# MapReduce example

Observations:

- The input pairs will be processed by different Map-workers
- Behind the scenes (invisible step), all emitted pairs with the same key are grouped together (after the Map phase and before the Reduce phase)
- The *grouping phase* includes concatenation of all the values corresponding to the same key
- In our example: in the grouping phase: three times $<"\text{komen}", 1>$ becomes $<"\text{komen}", [1, 1, 1]>$

# INTERMEZZO MapReduce example

Do you have any suggestions for optimization of the MapReduce program from the example on slide 12?

# MapReduce computing: early combining

- Word count could be optimized by doing some aggregation in the Map phase
- Instead of $k$ repetitions of $emit(< w, 1 >)$; do $emit(< w, k >)$;
- Adapt the Reduce program (how?)
- In general, this idea is applicable if the reduce function is commutative and associative (e.g. sum, max)
- Early combining often requires a setup of local datastructures and a final emit
- Our convention: for writing pseudo code, use functions $Init_Map()$ and $Finalize_Map()$

# Word count speed up

- Init_Map():
  Create a dictionary D (word, freq);
- Map $< docid, text >$:
  for each word $w$ in $text$
  add $w$ to D;
- Finalize_Map():
  for each entry (word, freq) in D
  $emit(< word, freq >)$;
- Reduce $< w, vlist >$:
  int $sum = 0$;
  for each $v$ in $vlist$
  $sum+ = v$;
  $emit(< w, sum >)$;

# Let's do it

Schrijf pseudocode voor Map en Reduce voor een collectie tupels van de vorm <g,v> die de volgende SQL query representeert:

```
SELECT g, SUM(v) FROM Input
WHERE v >= 100
GROUP BY g
HAVING SUM(v) >= 10000
```

# Let's do it again

Write pseudocode for Map and Reduce given a collection of tuples shaped <g,v> representing the following SQL query:

```
SELECT g, SUM(v) FROM Input
WHERE v >= 100
GROUP BY g
HAVING SUM(v) <= 10000
```

# MapReduce: references

- http://infolab.stanford.edu/~ullman/mmds/ch2.pdf